MCPA / SITETRACE

Cost-Benefit Analysis

Equipment accountability and inventory management

IT303 • Systems Analysis and Design
Prepared 24 October 2026 • Philippines

758 recorded units | 26 categories | 463 unreconciled units

Assessment status: Financial feasibility not yet determined. The inventory gap supports reconciliation and a measured pilot. It does not establish confirmed losses or a guaranteed return.

Companion workbook: SiteTrace_Cost_Benefit_Analysis.xlsx
Financial inputs are editable; this report is a dated baseline snapshot.

Contents

1. Introduction
2. Overview of the Existing System
3. Overview of the Proposed System
4. Existing Equipment Monitoring Analysis
5. One-Time Development Cost
6. Recurring System Costs
7. Tangible Benefits
8. Intangible Benefits
9. Existing versus Proposed System Comparison
10. Three-Year Financial Projection
11. Return on Investment and Payback Analysis
12. Interpretation of Results
13. Conclusion and Recommendations
14. Assumptions, Limitations, and References

1. Introduction

This cost-benefit analysis evaluates the proposed MCPA / SiteTrace equipment accountability and inventory management system against the existing manual or spreadsheet-based monitoring process. It is prepared for IT303 Systems Analysis and Design. The analysis uses the supplied equipment worksheet and a read-only review of the current repository. It does not rely on sample application records as company inventory evidence.

The supplied worksheet contains 758 recorded equipment units across 26 categories. Three categories have reported gaps totaling 463 units. These figures establish an accountability and reconciliation concern. They do not establish the cost of that concern or demonstrate that the computerized system is already financially feasible.

Table 1. Evidence and interpretation boundaries

Evidence class	Available evidence	Permitted interpretation
Supplied facts	26 categories; 758 recorded units; gaps of 19, 75 and 369	Observed worksheet quantities; not a dated physical count.
Reviewed capability	Equipment, movement, receipt, missing, repair and reporting source	Implemented code can support accountability; production effectiveness is not confirmed.
Planning conventions	3-month window, 3-year horizon, 8% discount input	Editable modelling assumptions; not actual labour effort or an approved hurdle rate.
Unknown finances	Rates, quotations, measured hours, spending and loss history	Financial totals, ROI, payback and NPV remain Insufficient Data.

2. Overview of the Existing System

The primary source is an undated worksheet titled “HAND TOOLS AND OTHERS MONITORING.” It records equipment descriptions, recorded quantities, need-to-comply quantities, and missing or no-report quantities. Only three categories have requirements and reported gaps. Blank cells mean that information was not supplied.

For comparison, the analysis models staff collecting counts, updating the worksheet, preparing reports, reconciling differences and following up with equipment holders. These activities are reasonable workflow inferences, not results of a staff interview or timing study. Staffing, reporting effort, reconciliation frequency, office supply costs and communication expenses remain to be measured. The product documentation specifies weekly monitoring, but it does not prove the actual calendar followed by the company. [R1, R2, R12]

3. Overview of the Proposed System

The reviewed system is a plain HTML, CSS and JavaScript web application with Supabase-backed data operations. The equipment masterlist stores identifiers, quantities, holders, sites and statuses. Shared movement code supports requests, approval and release, direct transfer, receipt inspection, returns, repair reports and missing-equipment recovery. Current authentication and permission code supersede older descriptions of prototype-only roles. [R4–R6, R9, R15]

Current report code distinguishes present inventory from saved historical snapshots and exports PDF reports. Movement activity also has PDF export. These current implementations take precedence over older CSV-only documentation. Snapshot capture is prospective and requires backend scheduling; the source does not establish that a production job is operating. Equipment movements select whole asset records, not arbitrary partial quantities. These limits affect the effort and benefits that should be measured during a pilot. [R6–R8, R11]

4. Existing Equipment Monitoring Analysis

Table 2 transcribes all 26 populated rows. A dash means not supplied, not zero. Descriptions are standardized for readability: “Water Pump” and “Catwalk Platform Scaffolding” retain the meaning of the source labels. The unpopulated row 27 is excluded. [R1, R2]

Table 2. Supplied inventory baseline

No.	Equipment description	Recorded	Need to comply	Missing / no report
1	Shovel (Pala, Lapad)	19	—	—
2	Shovel (Pala, Bilog)	23	42	19
3	Bareta	13	—	—
4	Itak	4	—	—
5	Piko	4	—	—
6	Plastic Drum 200L	14	—	—
7	IBC Water Tank 1000L	5	—	—
8	Wheel Barrow	4	—	—
9	Oxygen Tank	3	—	—
10	Gas Tank	5	—	—
11	Electric Kettle	2	—	—
12	Extension	10	—	—
13	Double Burner Stove	8	—	—
14	Lababo	5	—	—
15	Maso	4	—	—
16	Garden Hose	5	—	—
17	GI Pipes	225	300	75
18	Base Jack	27	—	—
19	U-Jack	56	—	—
20	Buckets	39	—	—
21	Catwalk Platform Scaffolding	33	—	—
22	H-Frame Scaffolding	64	—	—
23	Braces Scaffolding	41	—	—
24	Water Pump	2	—	—
25	Garden Sprinkler	2	—	—
26	Swivel Clamp	141	510	369
	Total recorded / provided cells only	758	852	463

4.1 Reconciliation interpretation

The 852 required units are a partial subtotal for the three flagged categories; no requirement was supplied for the other 23 categories. The total of 758 is the sum of recorded units of different equipment types, not a market valuation or a verified count of equivalent assets.

Table 3. Validation of the reported reconciliation gaps

Category	Requirement − recorded	Reported gap	Share of 463
Shovel (Pala, Bilog)	42 − 23 = 19	19	4.1%
GI Pipes	300 − 225 = 75	75	16.2%
Swivel Clamp	510 − 141 = 369	369	79.7%
Total	852 − 389 = 463	463	100.0%

All three reported gaps agree with requirement minus recorded quantity. Swivel clamps account for 79.7% of reported gap units, making that category a useful first reconciliation priority by volume. Unit volume does not establish financial priority because prices and specifications are unknown. The sheet does not distinguish unreported stock, items in transit, counting errors, unavailable items and confirmed physical losses. Adding the gaps to recorded stock would therefore not establish a confirmed company-wide stock total.

The workbook calculates exposure only when a flagged category has an entered replacement price. Potential value exposure equals reported gap units multiplied by the estimated price for the same unit and specification. This is a valuation of an unresolved position, not a confirmed loss, a promised recovery, or an annual benefit. The aggregate exposure remains Unverified until all three prices are supplied.

5. One-Time Development Cost

The workbook follows the academic “BREAKDOWN OF ONE TIME COST WORKSHEET” structure. The scope values full development and implementation effort. Because a system already exists, a future investment decision must distinguish remaining work from paid development already incurred. No retrospective payment, salary or development price is inferred from the existence of the code.

Table 4. One-time investment structure

Section	Included items	Calculation / current result
A. Development	System analysis; programming; database development; testing and QA; documentation and implementation	Rate per person-month × allocated person-months. Subtotal: Insufficient Data.
B. User training	Preparation; orientation; materials; training sessions	Rate per hour, kit or session × quantity. Subtotal: Insufficient Data.
C. Hardware / software	Deployment setup; separate database setup; backup setup; domain/SSL; necessary devices; licenses	One-time unit rate × quantity. Subtotal: Insufficient Data.
Total investment	Development + training + hardware/software	Insufficient Data. No actual company quotation supplied.

The three-month development window is an editable planning convention. It does not mean that every role works full time for three months. Rates and role effort are blank until supported. Separate preparation and delivery tasks should not be charged twice. Existing devices, open-source components, provider bundles and existing service capacity must be checked before assuming a new purchase is required.

> Total one-time cost = development cost + user training cost + additional hardware/software cost.

Each line has an editable treatment. Estimated and Confirmed require numeric inputs and an evidence basis. Bundled or Not applicable explicitly contributes zero. To confirm contributes Insufficient Data. A teaching example of ₱178,500 is not a SiteTrace budget and has not been imported into the analysis. [R3]

6. Recurring System Costs

The annual operating model includes maintenance, hosting, database charges, backup/storage, attributable internet costs, software/domain renewal, technical support and other necessary recurring services such as a scheduler. Provider usage, subscriptions, retention needs, available capacity and approved service allocations are not supplied. A Supabase-backed application does not by itself identify a paid plan or the cost of production operation.

Table 5. Recurring cost and double-counting controls

Cost group	Annual calculation	Boundary
Maintenance and technical support	Rate × quantity × annual frequency	Separate only when the services are distinct; bundled support is not repeated.
Hosting, database and backup	Rate × attributable quantity × annual frequency	Record provider bundles once. Free capacity is not presumed sufficient.
Internet, renewal and scheduling	Incremental rate × quantity × frequency	Include only costs attributable to SiteTrace. General existing connectivity is not charged in full.
Annual recurring total	Insufficient Data	Linked once to Financial Analysis.

Monthly and annual frequency defaults convert billing units; they are not subscription evidence. Residual administrative work and consumables belong in Proposed System Operating Costs. System hosting, maintenance and technical support belong in Recurring Costs. The academic example of ₱20,500 is excluded. [R3]

7. Tangible Benefits

7.1 Supplies and administrative cash savings

The comparison covers paper, printing/copy services, ink or toner, recordkeeping materials and other office supplies. Other administrative cash costs cover record storage and non-system reporting/communication. Existing and proposed rates, quantities and frequency are editable. Printing charges and ink must use distinct spending boundaries. Proposed office costs exclude system expenses already listed as recurring costs.

Supplies cash reduction = existing annual supplies cost − proposed annual supplies cost.

The reduction is attributable only if the system reduces an actual cash purchase, service charge or storage expense for the same workload. Reusing recorded inventory and PDF reports can reduce retyping and some printing, but these effects have not been measured. A decrease in paper use is a projection until supported by budgets or receipts and post-pilot use. Negative reductions remain additional costs; formulas do not force savings to be positive. [R7, R11]

7.2 Labour and productivity

The four modelled activities are inventory updates, report preparation, reconciliation and accountability follow-ups. Annual person-hours equal hours per person per event multiplied by staff and events per year. The proposed model uses the baseline reporting frequency and wage for comparability. Staff must include continuing data entry, inspection and physical checks. Activities need separate time boundaries so the same investigation is not counted twice.

Annual hours saved = existing annual person-hours − proposed annual person-hours.

Annual productivity value = annual hours saved × applicable hourly rate.

Faster work releases staff capacity, which may improve service without reducing payroll. The model therefore credits a default 0% of productivity value as cash savings. A higher cash-realization share needs a corresponding documented reduction in paid overtime, contractor hours or payroll expenditure. The retained portion is presented as noncash capacity value. Cash savings and retained capacity are never added twice. No employee dismissal or full-salary saving is assumed.

7.3 Equipment accountability and avoided procurement

The observed 463-unit gap gives a reason to improve reconciliation. The system can preserve named custody, inspect receipts, show missing cases and record recovery after verification. It cannot automatically locate or recover stock. No annual monetary benefit is calculated from the 463 units because confirmed loss history, prices, recurrence, avoidability and recovery evidence are absent. [R4–R6, R10]

Avoided replacement and avoided duplicate procurement are separate optional benefit claims. Both are excluded from the cash model by default. Inclusion requires annual evidence, a credible comparison with what would otherwise have been bought, and disjoint underlying events. A recovered historical item is a one-time event and must not be repeated as a yearly saving. Improved visibility may support purchasing decisions, but no automatic procurement optimizer or verified avoided purchase is claimed. [R14]

Table 6. Annual benefit classification at preparation

Measure	Current value	Treatment
Supplies and other admin cash reduction	Insufficient Data	Needs measured spending for both workflows.
Productivity / released capacity value	Insufficient Data	Needs task times, staffing, frequency and wage.
Payroll cash saving credited	₱0.00	Default exclusion, not a verified payroll expense.
Equipment / procurement saving credited	₱0.00	Default exclusion, not a finding that physical losses are zero.
Total annual cash tangible benefit	Insufficient Data	Cash benefit input to the main financial model.
Potential flagged stock exposure	Unverified	Separate from annual benefits; three replacement prices missing.

8. Intangible Benefits

The following benefits are supported by reviewed system features but cannot yet be assigned dependable monetary amounts. Their realization depends on correct deployment, complete records, staff use and continued physical verification. Suggested indicators provide a practical pilot evaluation plan.

Table 7. Qualitative benefits and feature support

Benefit	Supporting feature	Pilot measure
Equipment accountability	Named holder, reservation and inspected receipt associate an asset with an accountable person. [R4–R6, R10]	Share of held records with a named active custodian.
Visibility of assignments	Equipment profiles and project / personal lists expose recorded holder, quantity and location. [R4–R6]	Time to identify recorded holder and site.
Record accuracy	Eligibility verification and condition capture reduce inconsistent handovers. [R5, R6, R10]	Record-to-count variance and correction rate.
Faster equipment information	Search and filtered inventory views reduce repeated manual lookup. [R4, R7]	Median lookup time in matched pilot tasks.
Management decisions	Current inventory and movement reports can inform allocation and purchasing review. [R7, R11]	Percentage of purchase decisions checked against available inventory.
Fewer reporting errors	Report output reuses recorded fields and PDF export instead of retyping a separate summary. [R7, R11]	Report corrections per cycle.
Inventory transparency	Current state is separated from saved historical observations. [R7, R8]	Coverage of scheduled snapshots and reported exceptions.
Stronger audit trails	Saved movement actions, inspection notes and recovery actions support traceability. [R5, R6, R11]	Share of sampled handovers with complete movement / receipt evidence.
Management oversight	Role routes, missing / repair views and notifications expose unresolved work. [R6, R8, R9]	Age of unresolved discrepancy cases and overdue actions.

An audit trail is evidence of recorded actions, not proof of an actual physical count. Searchable data may still be stale. Historical snapshots do not reconstruct past inventory and need a configured backend job. Whole-record equipment handling also means that arbitrary partial receipts cannot be assumed to reduce workload. [R6–R8]

9. Existing versus Proposed System Comparison

Table 8. Operational comparison

Activity	Existing comparison workflow	Reviewed proposed support
Inventory monitoring	Spreadsheet quantity entries; dates/holder detail not supplied	Asset identifiers, quantity, site, holder and status views. [R4]
Movement accountability	Issue/return/transfer evidence not included in supplied sheet	Requests, reservations, release, named inspected receipt, return and recovery. [R5, R6]
Reconciliation	Compare requirements and recorded quantities; investigate gaps	Recorded movements and custody can help investigate; physical checks remain necessary. [R5, R10]
Report preparation	Manual/spreadsheet preparation assumed; effort unknown	Current and historical inventory views, movement history and PDF export. [R7, R11]
Financial cost	Admin cash and labour allocation unmeasured	Residual admin cash/labour plus recurring system expenses; totals unknown.

The proposed administrative cash cost is shown separately from recurring system cost. A full proposed cash view adds them once. Labour allocation is shown separately to explain resource use. The cash financial model uses the reduction in admin cash spending plus evidenced payroll and equipment benefits, then deducts recurring system cost. It does not subtract the entire proposed operating total a second time.

10. Three-Year Financial Projection

The model has an initial investment at Year 0 and three year-end operating cash flows. The 8% annual discount rate is an adjustable modelling convention. Benefit-realization and recurring-cost multipliers default to 100% for each year and can be changed independently. These are scenario controls, not an adoption forecast. The baseline report uses no invented prices or salaries.

Table 9. Cash projection at the supplied-data baseline

Measure	Year 0	Year 1	Year 2	Year 3
Cash tangible benefits	₱0.00	Insufficient Data	Insufficient Data	Insufficient Data
Recurring system costs	₱0.00	Insufficient Data	Insufficient Data	Insufficient Data
Initial investment	Insufficient Data	₱0.00	₱0.00	₱0.00
Net cash flow	Insufficient Data	Insufficient Data	Insufficient Data	Insufficient Data
Cumulative cash flow	Insufficient Data	Insufficient Data	Insufficient Data	Insufficient Data
Discount factor	1.0000	0.9259	0.8573	0.7938
Discounted net flow	Insufficient Data	Insufficient Data	Insufficient Data	Insufficient Data

Year 0 operating amounts and later one-time investment entries are zero by model timing, not evidence of free operations. Cash tangible benefits consist of admin cash reductions and any separately supported payroll and equipment benefits. Costs already present in the comparison are not deducted again.

Annual net benefit = annual cash tangible benefits − annual recurring system cost.

NPV = −initial investment + Σ[annual net benefit in year t / (1 + discount rate)^t], t = 1 to 3.

11. Return on Investment and Payback Analysis

Table 10. Financial results and calculation basis

Indicator	Calculation basis	Baseline result
Three-year ROI	(Sum of Years 1–3 net operating benefits − investment) / investment	Insufficient Data
Discounted benefit-cost ratio	PV of cash benefits / (initial investment + PV of recurring costs)	Insufficient Data
Payback period	Investment / constant positive annual net benefit; otherwise first cumulative crossing	Insufficient Data
Cash NPV	Initial outlay plus discounted operating cash flows	Insufficient Data
Productivity-inclusive economic NPV	Cash NPV plus PV of retained capacity value; not cash flow	Insufficient Data

Payback interpolation assumes benefits accrue evenly within each year. If the investment is not recovered within three years, the workbook reports that outcome explicitly. A zero investment produces a defined non-applicable ROI rather than division by zero. The benefit-cost ratio is discounted consistently with NPV. This makes the incremental comparison explicit rather than suggesting that all existing company costs are system benefits.

The workbook also computes the level annual cash benefit needed for zero NPV: annual recurring cost plus investment divided by the three-year discount-factor sum. For simple recovery in three years, the requirement is recurring cost plus one third of investment. These thresholds remain Insufficient Data until investment and annual recurring cost are known. They help assess a quotation without inventing a benefit forecast.

12. Interpretation of Results

The inventory evidence supports the operational need for better reporting and reconciliation. All three gap calculations match the supplied sheet. However, a gap is not a financial loss statement. The missing information prevents a reliable appraisal of the peso exposure, annual savings, implementation cost and recurring cost. The financial conclusion is therefore Not Yet Determined; the current data does not justify an affirmative or negative ROI claim.

The code review gives a credible mechanism for benefit: named custody and inspection evidence can make follow-up more focused; current inventory and PDF export can reduce duplicate reporting; and saved history can help explain recorded changes. These are capabilities, not measured savings. No code-only review can establish user compliance, actual deployment, production reliability, successful recovery or a reduction in payroll and purchasing expenditure.

Productivity and cash feasibility can lead to different conclusions. A system could improve reporting and release staff time while still requiring additional cash for hosting and maintenance. The supplemental economic NPV makes that distinction visible. It must not be reported as cash return. Similarly, a high replacement value exposure does not make the whole exposure a realizable yearly benefit.

13. Conclusion and Recommendations

SiteTrace has reviewed features that are relevant to the worksheet’s accountability problem. The evidence is sufficient to recommend a controlled reconciliation and operational pilot. It is insufficient to approve a financial return or guarantee recovery of 463 units. A final investment recommendation should follow completion of the workbook inputs and review of the resulting cash projection.

• Reconcile the three flagged categories. Confirm specifications and units, count at relevant sites, identify holders, inspect issue/transfer/return records and approve any adjustment with evidence. Classify unresolved items before recognizing a physical loss.

• Measure a representative baseline and matched pilot, suggested over four weekly reporting cycles. Record separate person-hours for monitoring, report preparation, reconciliation and follow-up. Include continued data entry and physical checking in proposed effort.

• Obtain a scoped remaining-work quotation and record any historical development spend separately. Confirm person-month allocations, training needs, existing devices and provider bundles. Do not copy another project’s teaching budget.

• Enter actual or approved estimated wages, supplies purchases, incremental connectivity, hosting, backup, maintenance and support. Record dates, source documents and whether each cost is estimated, confirmed, bundled or inapplicable.

• Credit payroll reduction only when an expense actually declines. Support equipment and duplicate-procurement claims with distinct annual events and avoid repeating a historical recovery as a recurring benefit.

• Confirm production authentication, server authorization, availability verification and scheduler readiness as part of implementation planning. Recalculate the workbook, test conservative benefit-realization scenarios, and update the Word/PDF snapshot before seeking investment approval.

14. Assumptions, Limitations, and References

14.1 Assumptions and limitations

The analysis is dated 24 October 2026 in Asia/Manila. The inventory worksheet has no supplied observation date, signed count, valuation, individual custody listing or reconciliation outcome. Requirements for 23 categories, actual salaries, task timings, staffing, supplies expenditure, service quotations, confirmed losses and avoided purchases were not supplied. No live database was queried and no production configuration was changed.

Table 11. Modelling assumptions and exclusions

Item	Treatment
Development duration	3 calendar months as an editable planning window; actual person-months remain unknown.
Discount rate and horizon	8% editable convention, 3-year fixed projection; no claim that 8% is an approved company rate.
Workload and wages	Proposed frequency and hourly rates link to existing baseline; staff/time remain editable.
Timing / escalation	Year-end cash flow; within-year payback interpolation; constant-price defaults; editable year multipliers.
Other financial effects	No tax shield, financing cost, terminal resale value or monetary theft-prevention benefit credited. Use consistently defined quotations.
Exclusion defaults	0% payroll realization and excluded equipment claims are model policies, not verified zero expenses or zero losses.
Source priority	Current implementation takes precedence over old feature prose; live deployment and measured performance remain unverified.
Report and workbook	Report is a static baseline snapshot. Editing Excel does not update Word/PDF automatically. Rebuild or revise both together.

All peso financial inputs remain unknown unless a documented value is entered. No external equipment price or hosting price has been substituted for company evidence. The source photographs contain another project's sample amounts of ₱178,500 and ₱20,500; these are excluded from SiteTrace totals. The photographs supply the academic layout only. [R3]

14.2 References and source register

References are supplied primary materials and repository sources reviewed on 24 October 2026. The academic photographs do not provide a complete author, institution, publication year or textbook title. No missing bibliographic detail has been invented. Repository line anchors, hashes and revision are recorded in validation/source-evidence.json.

[R1] Supplied inventory image. HAND TOOLS AND OTHERS MONITORING; undated; 26 populated rows. Blank row 27 excluded.

[R2] User-supplied transcription. Pasted text.txt; supplied quantities cross-checked against R1.

[R3] Academic reference images. IT303 Systems Analysis and Design, Unit 2; visible printed page 105 and subsequent benefits worksheet / viewer pages 58–59. Layout only; full bibliographic details not supplied.

[R4] Asset masterlist. 1-admin/modules/masterlist/masterlist.js. Registration, quantity, site, holder and condition/status views.

[R5] Authenticated movement client. js/movement-cloud.js. Requests, release, transfer receipt, returns, repairs and recovery actions.

[R6] Movement interface. js/movement-ui.js. Whole asset records; per-tool receipt inspection and follow-up cases.

[R7] Reports implementation. 1-admin/modules/reports/reports.js. Current inventory, prospective snapshots, movement reports and PDF export.

[R8] Monitoring backend. supabase/migrations/202610100006_monitoring.sql. Prospective weekly snapshot functions; scheduler remains external.

[R9] Permissions and identity. js/permissions.js. Role routes/actions; authenticated Supabase sessions.

[R10] Accountability verification. supabase/migrations/202610200001_verified_request_inventory.sql. Server reservation verification and availability checks; migration presence does not prove deployment.

[R11] Activity log export. js/movement-overview.js. Saved movement history with PDF export.

[R12] Product operating context. PRODUCT.md. Weekly Monday 16:00 monitoring requirement. Planning prose can be stale; code takes precedence.

[R13] Current materials scope. documentation/consumables-upgrade-2026-10-22/REPORT.md. Consumables have separate quantities, receipts and stock history; illustrative seeds excluded.

[R14] Purchase ledger scope. 1-admin/modules/purchases/purchases.js. Ledger/reference evidence and private receipt files; no evidence of avoided procurement.

[R15] Authentication source. js/auth.js. Actual sign-in/session controls supersede older prototype-only descriptions.

14.3 Workbook completion guide

Start with Assumptions and References. Complete blue inputs and choose a treatment for every cost line. Enter measured task data for both workflows, document supplies and administrative cash costs, and keep equipment claims excluded until supported. Inventory replacement prices are optional exposure inputs. Review Tangible Benefits, Financial Analysis and Executive Dashboard after recalculation. Confirm negative values, no-recovery results and missing-data messages before updating this report.